Milind Chauhan

Software Engineer - AI/ML
Microcrispr Pvt. Ltd.

+91-8989550808
milind05.chauhan@gmail.com
LinkedIn Profile
GitHub Profile

Summary

Software Engineer (AI/ML) with hands-on experience building end-to-end AI/ML systems spanning predictive modelling, deep learning, LLM fine-tuning, and production-grade MLOps with robust backend services and interactive web applications . Proficient in PyTorch, TensorFlow, and agentic coding workflows (Claude Code-style iteration). Experienced in designing offline evaluation harnesses, experiment tracking, model versioning, and reproducible pipelines on AWS. Strong foundations in algorithms, probability/statistics, and translating complex real-world problems into measurable ML formulations. Background in computational biology brings domain depth in sequence modelling and protein language models.

Experience

- **Software Engineer - AI/ML** *Jan 2025 – Present*

  Microcrispr Pvt. Ltd. New Delhi

  - Designed and maintained scalable data ingestion and transformation pipelines using AWS Glue, processing multi-source datasets (10^4–10^5 records per run) with access control and metadata governance via AWS Lake Formation.
  - Developed serverless, event-driven microservices using AWS Lambda to automate data preprocessing and model-triggered workflows, reducing pipeline turnaround time by ∼40–50%.
  - Implemented an Apache Kafka–based streaming layer for asynchronous communication between distributed analytics services.
  - Developing a full-stack platform for drug discovery and prioritization, deploying 5+ containerized microservices on AWS EKS with modular service design and fault-tolerant communication via Kafka, including a high-performance knowledge graph visualizer for 1.1M nodes and 6M edges backed by Neo4j, with interactive graph exploration interfaces.
  - Developing a U-Net-based super-resolution pipeline for histopathology, reconstructing high-resolution images from low-resolution inputs and achieving a validation PSNR of 34.94 on held-out samples. Github.
  - Maintained experiment tracking and model versioning using MLflow and Weights Biases; managed containerised deployments via Docker and Kubernetes (EKS) with CI/CD pipelines — directly supporting reproducible, production-ready model rollouts.

Technical Skills

Programming: Python, C++, R
Fullstack & Backend Development: Javascript, Typescript, FastAPI, React, Node.js, Next.js, RESTful APIs, HTML5, CSS3, Bootstrap, PHP
Tools & Frameworks: PyTorch, TensorFlow, Scikit-learn, OpenCV, AutoGluon, QuPath
Data Systems & Databases: PostgreSQL, MySQL, MongoDB, Neo4j, Redis, DuckDB, Vector Databases (ChromaDB)
Cloud & DevOps: Git, Docker, AWS (EC2, S3, SageMaker, Lambda, Glue, Lake Formation), Kubernetes (EKS), Apache Kafka, CI/CD, Container Orchestration
GenAI & LLMs: OpenAI APIs, Gemini, Llama, HuggingFace Transformers, LoRA/QLoRA, Model Fine-Tuning, RAG Pipelines, Agno, LangChain
MLOps: MLFlow, Weights & Biases, Model Deployment, Logging & Monitoring, Model Versioning, Experiment Tracking, GPU Optimization

Projects

- **AI-Assisted Molecular Dynamics (MD) Analysis and Reporting Workbench**

  Full-Stack ML Platform | Local RAG

  - Designed and built an installable, end-to-end molecular dynamics (MD) analysis platform that converts raw simulation trajectories into reproducible analyses, publication-quality visualizations, and interpretable scientific summaries.
  - Automated standard post-simulation workflows (RMSD, RMSF, radius of gyration, hydrogen bonding, PCA, contact maps, SASA) using established MD libraries, eliminating manual scripting while ensuring deterministic and auditable results.
  - Implemented and integrated a locally deployed Retrieval-Augmented Generation (RAG) system with a publication-oriented reporting module to generate literature-grounded interpretations, journal-style plots, figure captions, and draft Results sections, reducing analysis and reporting time by >70%.
  - Technology Used: Python, RAG, VectorDB, Local-LLM, ElectronJS

- **AntiCP3**
  Prediction of Anticancer Proteins using Large Language Models (LLMs)
  - Developed a novel computational framework for anticancer protein prediction by fine-tuning a state-of-the-art protein language model (ESM2). Deployed as HuggingFace and Github Repository, Pip package, standalone CLI tool and web interface. Available here.
  - Technology Used: TensorFlow, scikit-learn, Transformers(HuggingFace)

Education

- **M.Tech. in Computational Biology** *2023-25*
  IIIT-Delhi CGPA: 9.12
- **M.Sc. in Bioinformatics** *2021-23*
  Bioinformatics Centre, SPPU CGPA: 8.85

Research Publications

- **Chauhan M.**, Gupta A., Tomer R., and Raghava GPS. "CancerPPD2: an updated repository of anticancer peptides and proteins" *Database*, Volume 2025, 2025, baaf030, May 2025.
- Gupta A., **Chauhan M.**, Tomer R., and Raghava GPS. "AntiCP3: Prediction of Anticancer Proteins Using Evolutionary Information from Protein Language Models" *BiorXiv*.

Milind Chauhan
Date: March 28, 2026